

High Amplitude Lift Tracking Using Closed-Loop Feedback and Control: A Flow Analysis

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Abstract

The flow field around a C_L -controlled harmonically pitching flat plate was characterized using Particle Image Velocimetry (PIV). Analysis of the flow field, namely the growth and evolution of the leading-edge vortex (LEV) and other flow features were compared for a C_L -controlled pitching flat plate across different reduced frequencies and C_L amplitudes. A closed loop control architecture using a PID controller was used for the present study which resulted in non-harmonic angle of attack histories to achieve a harmonic lift coefficient reference. This analysis showed that although drastically different flow features were present across the flow regimes tested, the PID controller was able to linearize the dynamics and perform adequate tracking of the harmonic reference signals.

Nomenclature

α	= geometric angle of attack (deg)	ω	= angular frequency (rad/s)
α_d	= desired geometric angle of attack (deg)	f	= frequency (Hz)
c	= chord	t	= time (s)
C_L	= lift force coefficient	τ	= convective time
s	= wing positing with respect to gust	E_{AVG}	= average error
U_∞	= freestream velocity (m/s)	A	= C_L amplitude
$\dot{\alpha}$	= pitch rate (rad/s)	Abbreviations	
$\ddot{\alpha}$	= pitch acceleration (rad/s ²)	GR	= Gust Ratio
b	= semi-chord (m)	PID	= Proportional Integral Derivative
a	= pitch axis location in semi-chord (m)	FSF	= Full State Feedback
$C(k)$	= Theodorsen's function	KF	= Kalman Filter
$\dot{h}$	= plunge rate (m/s)	LPF	= Low Pass Filter
$\ddot{h}$	= plunge acceleration (m/s ²)	LEV	= Leading Edge Vortex
k	= reduced frequency		

I. Introduction

The impetus for the work presented in the paper is based on the results of Mongin et al. [1] where several closed loop control system architectures were developed in the University of Dayton Water Tunnel to control the C_L of a pitching wing at various tracking amplitudes and frequencies, to simulate a sort of 'virtual gust'. The goal of this

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study was to develop a control system to maintain C_L in external gust encounters. The study used three different types of controllers: a proportional integral derivative controller (PID), a full state-feedback controller (FSF), and a full state-feedback controller with Kalman Filter (FSF-KF). The three controllers were evaluated with Bode plots to assess their bandwidth across a range of reduced frequencies and amplitudes of the reference coefficient of lift signal. Across the three controllers it was observed that for similar harmonic reference signal tracking (phase and amplitude alignment) there were differing non-harmonic angle of attack histories. This was attributed to the different ways the controllers treated their tracking error. For instance, the PID controller was tuned with a higher gain on the integral portion of the error than the proportional or derivate gains, resulting in an aggressive response as error in the tracking grew over time. The FSF and FSF-KF controllers used C_L and its derivative as the feedback states and did not exhibit an aggressive response with growth in tracking error. One of the most prominent instances of this is shown in Fig. 1 where a reference signal of reduced frequency ($k = \frac{\omega c}{2U_\infty}$) of 0.01 and coefficient of lift amplitude (A) of 1 is shown.

This case is of particular interest because the reduced frequency is considered quasi-steady, meaning unsteady effects are minimal or not present, and the C_L amplitude is larger than the maximum static C_L of 0.8. Fig. 1(A) shows the C_L as measured on the flat plate from the three types of controllers plotted with the target or reference signal. Fig. 1(B) shows the angle of attack histories of each controller time aligned with the previous plot. In this case the PID controller maintained a better tracking to the target C_L throughout the entire profile. The FSF controllers performed less accurate C_L tracking, specifically during regions where the C_L signal exceeded 0.8 which can be seen clearly in Fig. 1(A). The angle of attack profile shown in Fig. 1(B) of the PID controller was noticeably different than the angle of attack profiles of the FSF controllers.

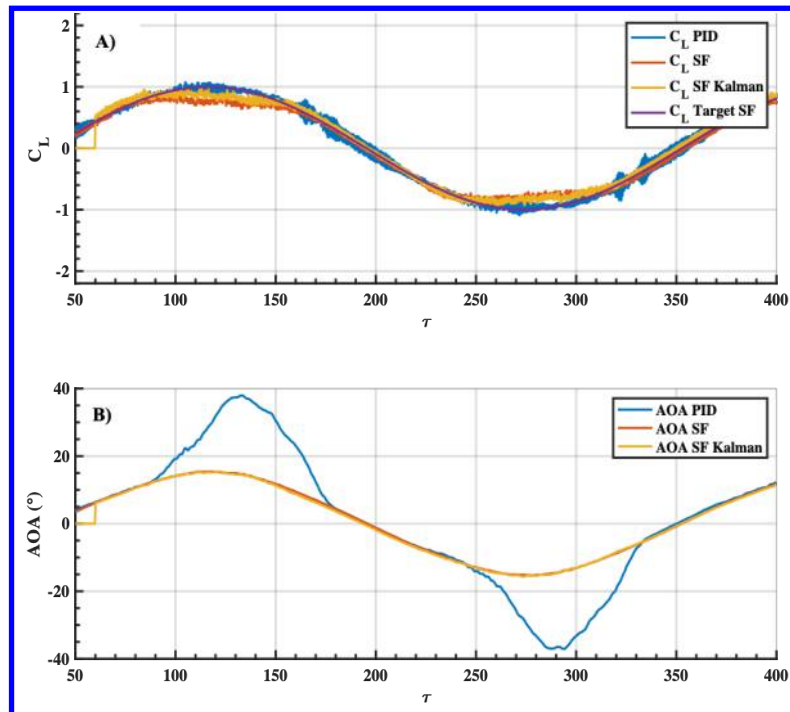


Fig. 1 Closed Loop Results for $k = 0.01$ $A = 1$ for Coefficient of Lift (A) and Angle of Attack (B) [1]

Differing angle of attack profiles were present through many C_L frequency and amplitude cases in the study and an understanding of the cause was difficult to assess without the knowledge of the flow field. The results from the study however, showed a capability of the closed loop system to track coefficient of lift reference signals up to amplitudes of 2.8 and reduced frequencies up to 3. The results of the PID controller for the $k = 0.2$ and $k = 0.3$ cases are shown in Fig. 2 plotted as coefficient of lift versus angle of attack for each coefficient of lift amplitude denoted by the colors. As reduced frequency was increased to $k = 0.3$, the C_L loops show different trends and behavior when compared to $k = 0.2$ case as annotated in Fig. 2. For $k = 0.2$, the pitch up and pitch down “routes” collapsed for different C_L amplitude cases whereas the routes don't collapse in $k = 0.3$ case. During the mid-region of travel the $C_{L\alpha}$ slopes mimicked that of the static lift curve however, above the static maximum coefficient of lift region, the paths

deviated greatly from the static curve. The behavior in these higher angle of attack regions, especially during the pitch down motion is suspected to be related to the leading-edge vortex (LEV) and trailing edge vortex (TEV) formation and shedding.

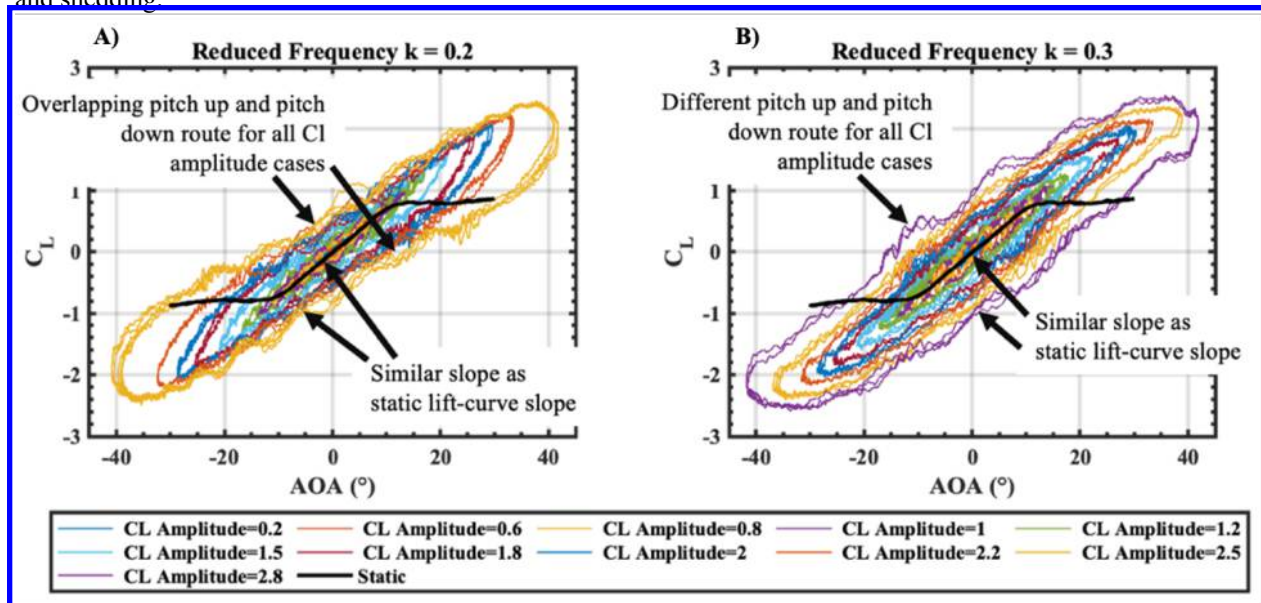


Fig. 2 C_L vs α PID at $k = 0.2$ (A) and $k = 0.3$ (B) [1]

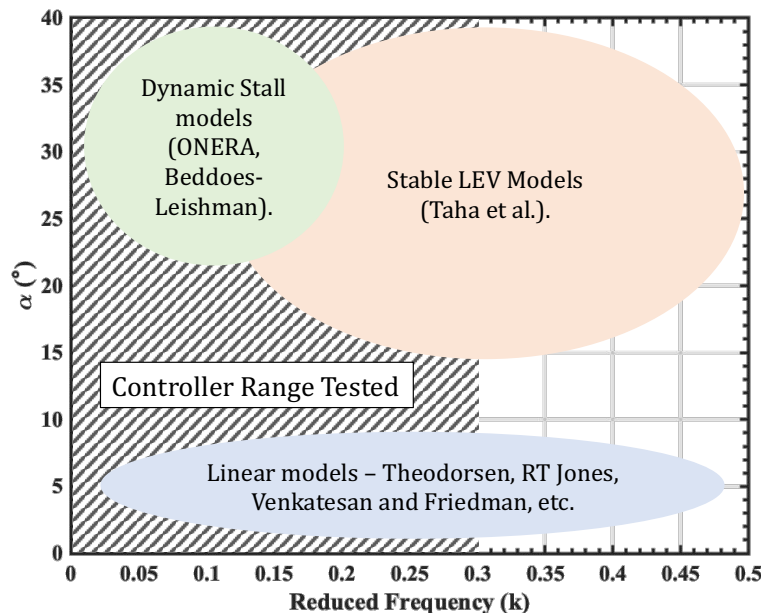


Fig. 3 Reduced Frequency vs Magnitude of Angle of Attack Oscillation Overlay of Known Models and the Current Study

Fig. 3 shows a range of reduced frequencies and angles of attack that the designed controller operates in along with various models that predict the C_L response from the wing at varying levels of accuracy. The controller operates in the linear region where Theodorsen's [4] unsteady model, RT Jones model [5], and Venkatesan and Friedman model [6] are historically used at lower angle of attack ranges where the boundary layer is presumed to be fully attached. The controller also operates in the higher angles of attack, low reduced frequency range where unstable leading-edge vortex is known to form and burst leading to a dynamic stall behavior. Models such as ONERA [7], and

Beddoes-Leishman [8] models are used in these ranges. The controller also operates in the highly unsteady range at higher amplitudes where a stable LEV forms leading to significant contributions in the lift. Taha et al. [9] model captures the effect of lift due to LEV. Different plant or state-space models were developed in these regions to capture the flow dynamics. However, the controller built in Mongin et al. [1] is purely sensor based and doesn't have the knowledge of the aerodynamics involved in the different zones shown in Fig. 3.

II. Experimental Setup

All experiments were performed at the University of Dayton Water Tunnel, shown in Fig. 4a. For a detailed explanation of the experimental setup as well as calibration of the tunnel speed and turbulence intensity refer to Mongin et al. [1]. The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. For this study, the freestream velocity of the tunnel was set to 0.28m/s and a flat plate with a 3-inch (0.076m) chord and 6-inch (0.152m) span was used. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinion gear that converts its linear motion to a rotational motion in order to pitch a test article in the tunnel test section as shown in Fig. 4b.

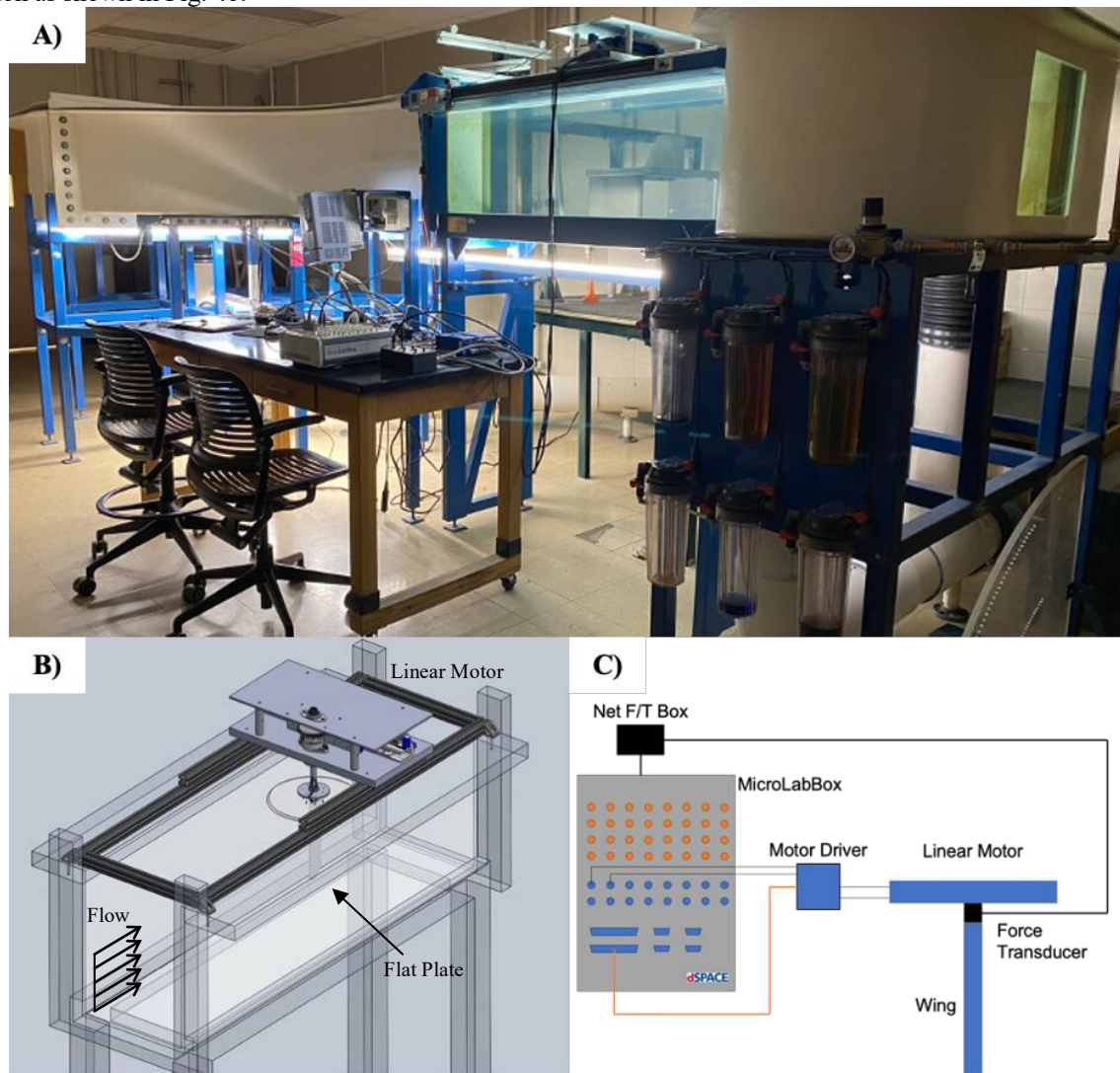


Fig. 4 University of Dayton Water Tunnel facility (A) CAD schematic of gust generator and test article (B) wiring schematic of MicroLabBox and input/output devices (C)

The closed-loop feedback and control for this system is performed with a dSPACE MicroLabBox with BNC type terminations using the MATLAB LabVIEW Real Time Interface (RTI) programming environment. The MicroLabBox is capable of up to 24 channels of analog input and 16 channels of output in 16-bit at 1 Mbps. There are also 48 channels of bidirectional digital inputs and outputs available. The MicroLabBox was run at 1kHz for these experiments although it is capable of up to 2Ghz processing in real-time. Force measurements were performed with an ATI Mini40 force transducer with NET F/T box interfaced with the dSPACE equipment using the ethernet protocol blocks in Simulink. A wiring schematic of the feedback system is shown in Fig. 4c. The force transducer F/T outputs are taken into the MicroLabBox and used to calculate aerodynamic forces on the test article. The motor driver is connected to two analog outputs, one for enabling the motor and the other for commanding the current of the motor. Another connection between the motor driver and MicroLabBox is made via the digital I/O ports to receive information from the motor encoder.

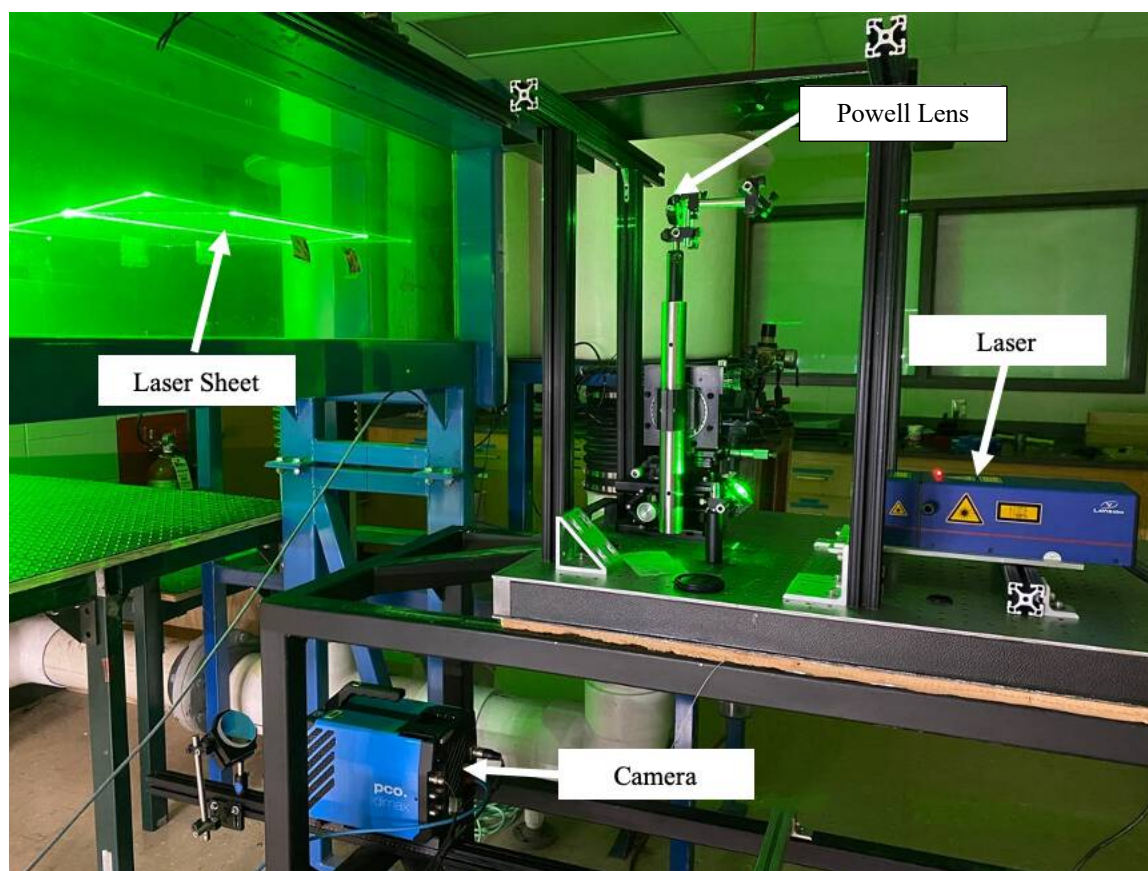


Fig. 5 PIV Camera, Laser, and Optics System

The PIV system shown in Fig. 5 is comprised of a PCO dimax S4 high speed camera with a 2016x2016 (4 Megapixel) CMOS sensor and a 3-Watt continuous wave laser from LaVision. The field of view is oriented from the bottom of the test section facing up, requiring the camera to pass through a 45-degree mirror. The laser beam is passed through a series of mirrors and into an Edmunds Scientific Powell lens and projected as a light sheet with a 30 degree fan angle. 60 μ m polyamide seeding particles from LaVision are used in the tunnel to be illuminated by the laser. Triggering of the camera recording is done with the MicroLabBox and the trigger signal is recorded along with the time locked data from the sensor and the pitch position of the test article. This ensures that the PIV data and the MicroLabBox data are time synced. Post processing of the PIV image frames is done with the Fluere software from Kyle Lynch [3]. The post processed PIV data and the data collected from the MicroLabBox are combined and analyzed using MATLAB.

The block diagram of the closed loop control system schematic shown in Fig. 6 displays the generalized

layout of the control system. It is comprised of a coefficient of lift controller which receives the coefficient of lift error from the difference between the desired coefficient of lift (C_{L_d}) and the sensed coefficient of lift (C_L) and determines a desired angle (θ_d) for the system to achieve. This is then received by the motor and wing controller that commands the wing to that angle to achieve the desired C_L measured by ATI Mini40.

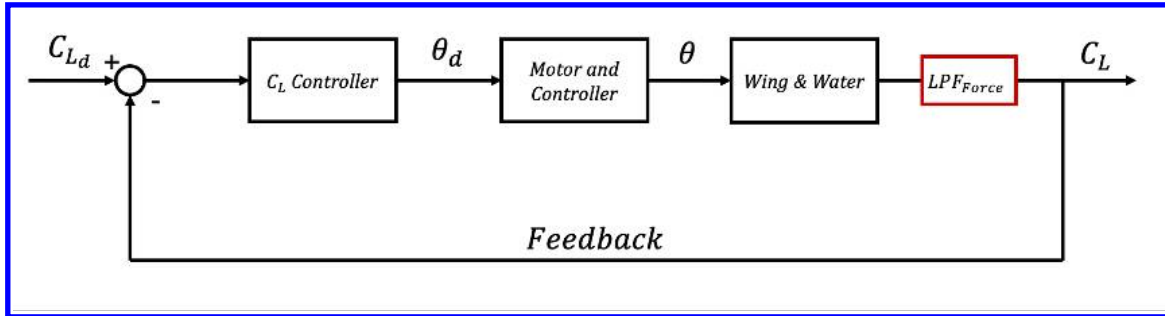


Fig. 6 Block Diagram of System Containing Motor Controller and Coefficient of Lift Controller

III. Results

The static C_L response to fixed α is shown in Fig. 7. The static lift curve was obtained by changing the angle of attack one degree at a time and recording forces for 10 seconds from -30 to 30 degrees and from 30 to -30 degrees and averaging the two directions together. From these tests a maximum C_L of 0.8 is assumed for the flat plate at static angles. The test matrix shown in Table 1 was completed for the closed loop cases with a PID C_L controller. Included in the test matrix are the tracking error results from the tests performed. The limited range of C_L amplitudes tested for the $k = 0.1$ cases was due to the controller's tendency to command extremely large angles of attack (great than ± 50 degrees) to achieve C_L amplitudes greater than 1.5 at this reduced frequency. The limitation of ± 50 degrees was applied across all tests to protect the equipment in the experimental setup and ensure operation of the closed loop controller in a stable and non-divergent range. The tracking error was calculated using Equation 1. The tracking error increases with the increase in the reduced frequency and the C_L amplitude. The maximum errors were seen in the highest C_L amplitude for both 0.2 k and 0.3 k cases with over 20% error.

$$E_{avg} = \frac{1}{n} \sum |C_{L_{target}} - C_{L_{sensed}}| * 100 \quad (1)$$

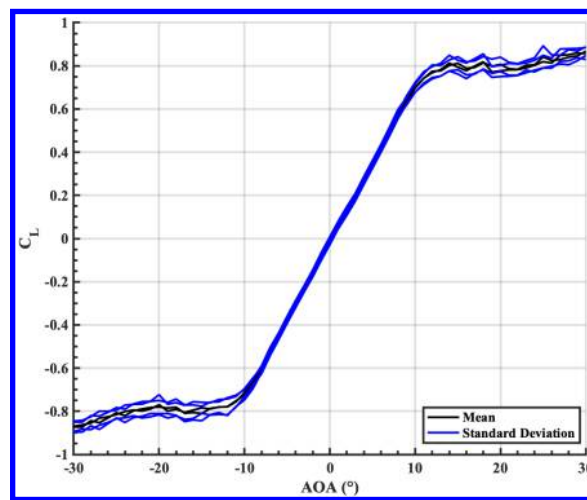


Fig. 7 Open Loop Static Coefficient of Lift

Table 1 Test Matrix for the closed loop control experiment.

Reduced Frequency	Sinusoidal C_L Amplitude	E_{avg}
$k = 0.1$	0.5	1.53
	1	2.95
	1.5	6.98
$k = 0.2$	0.5	2.91
	1	3.84
	1.5	5.94
	2	10.72
	2.5	28.53
$k = 0.3$	0.5	5.87
	1	6.62
	1.5	9.04
	2	13.51
	2.5	23.59

Fig. 8 (A) shows the C_L controlled results for $k = 0.1$ case at various amplitudes shown in Table 1. The lowest amplitude of 0.5 is quite linear and displays minimal hysteresis whereas the 1 and 1.5 amplitude cases begin to show large hysteresis as phase lag between the actuation and the lift response increases, especially in the 1.5 amplitude case where the angle of attack reach upwards of 20° where dynamic stall occurs (Fig. 3). As such, with increase in amplitude, the shape of the C_L vs α profile changes, primarily in the areas where the plate transitions from pitch up to pitch down maneuver. In these transition regions, the profile appears skewed and less symmetric than in the lower amplitude cases. The plot in Fig. 8 (B) shows the C_L with the added mass component [4] (shown in Equation 2) subtracted from the sensed C_L .

$$C_{LAM} = \pi b \left(\frac{\dot{\alpha}}{U_\infty} - \frac{ba\ddot{\alpha}}{U_\infty^2} \right) \quad (2)$$

As expected, from Fig. 8 (B) the added mass contribution is at its highest at low α where $\dot{\alpha}$ is at its highest. During the transitions where the $\dot{\alpha}$ is close to zero, the hysteresis or the skewness appear to be due to the circulatory component around the flat plate. At lower C_L tracking amplitudes, the circulatory lift coefficient (C_L^C) appear to be very similar and almost lay on top of one another in the regions from -5 to 5° . Whereas at higher angles of attack ($> 20^\circ$) the trends in the hysteresis loop changes as seen in the 1.5 CL amplitude case.

The velocities and resulting vorticity throughout the C_L tracking was captured using PIV. The flow field around the flat plate during the closed loop maneuver is shown in Fig. 9 along with the convective time history of C_L and α for the $k = 0.1$ case at a C_L amplitude of 1.5. The commanded C_L of 1.5 was able to be achieved due to a large recirculation region, represented as positive vorticity, formed on top of the plate that covers the entire length of the chord as seen in Fig. 9. However, in the transition region from pitch up to pitch down, the controller required a larger angle of attack range from 18° to almost 30° to track the peak C_L magnitude between 1 and 1.5. Since the C_L is seen to lead the angle of attack, once the peak C_L was achieved, the controller reduced the $\dot{\alpha}$ but continues to increase α until C_L decreased. This could be attributed to the similarities in the vorticity around the plate seen at the convective times of 5, 8, and 10 where the magnitudes of C_L were fairly constant even as the angle of attack changed by roughly 12° .

At the point of maximum α (and 0 $\dot{\alpha}$), the vorticity appears to begin to convect away from the plate. As the plate continues to pitch downward and closer to 0 α , the flow field returns to a state of balance between the intensities of vorticity on the upper and lower surfaces and the C_L approaches 0. The circulatory lift component C_L^c shows a minimal magnitude difference from the net C_L since the added mass component being small at this reduced frequency.

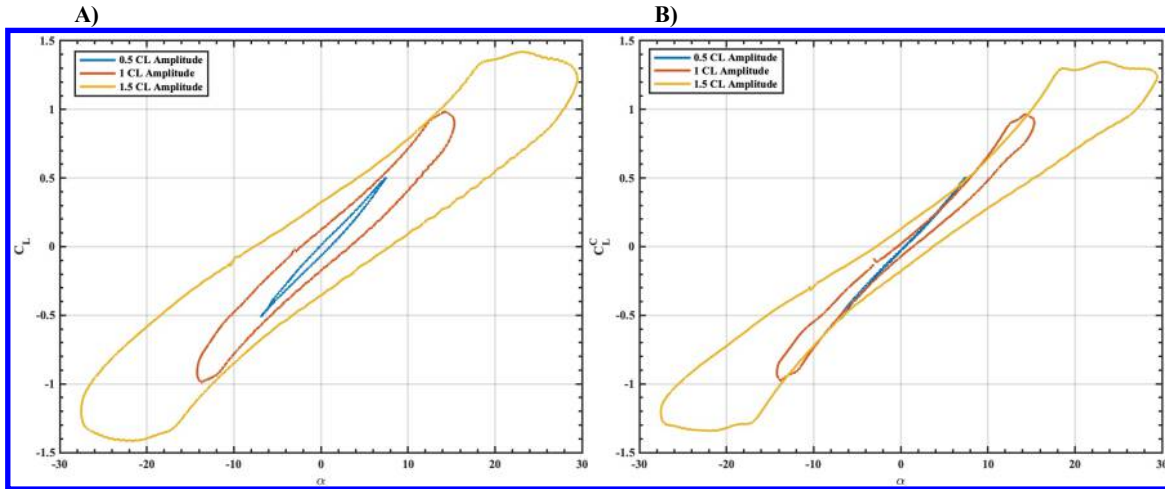


Fig. 8 Closed Loop C_L vs Angle of Attack for $k = 0.1$ (A) and C_L^c vs Angle of Attack (B)

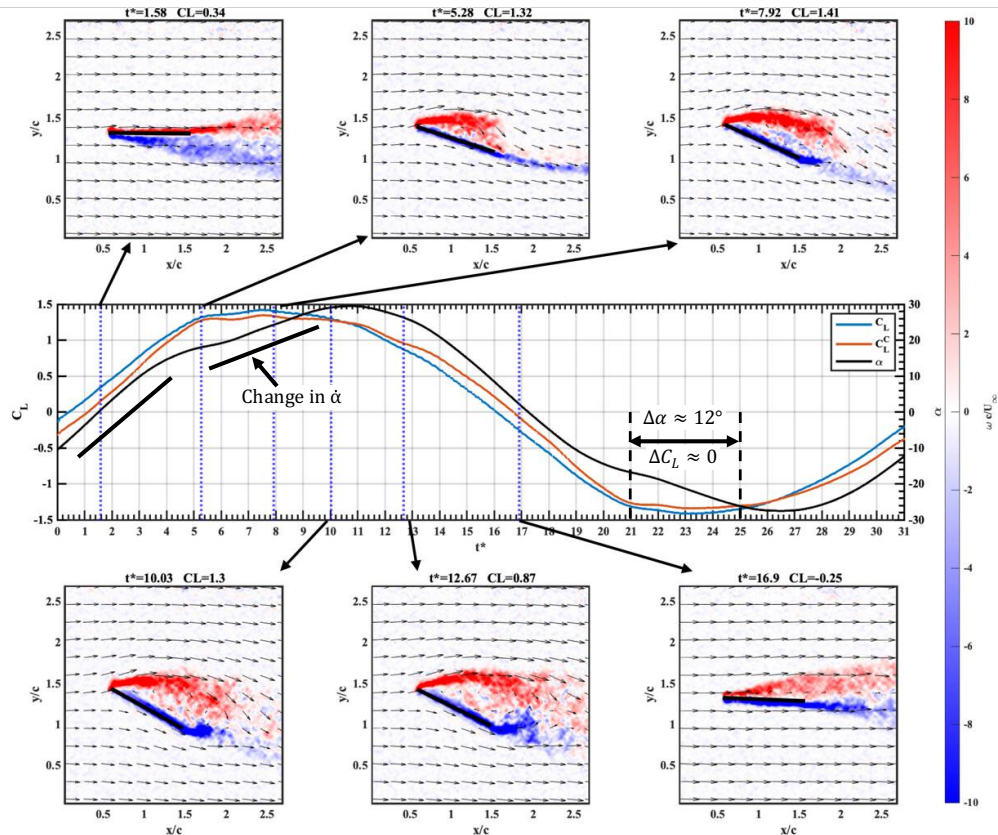


Fig. 9 Flat Plate Closed Loop Profile ($k = 0.1$ $C_{LAMP} = 1.5$) of C_L , C_L^c , α , and Select PIV Frames

Similar results are shown in Fig. 10 for $k = 0.2$ which is on the threshold of the unsteady and highly unsteady regime where the leading edge vortex contribution play a dominant role in the overall magnitude of the lift generated by the wing. Significant differences are seen in the transition regions between $k = 0.1$ and $k = 0.2$ cases. Unlike in $k = 0.1$ case where large angle of attack is required to maintain constant C_L , a drastic decrease in C_L magnitude is seen from 28° to 30° angles of attack in the $k = 0.2$ case. This could be attributed to the formation and shedding of the leading-edge vortex as shown in Fig. 11 which is not fully established in the $k = 0.1$ case. The behavior of the controller during these cases appeared similar throughout the different C_L amplitudes. The circulatory lift coefficient shown in Fig. 10 (B) shows a different behavior when compared to k of 0.1. There are defined ‘cusps’ which is caused by the delay in the transition from the pitch up to the pitch down maneuver. Similar to 0.1 k case, since the C_L leads α , once the peak C_L is reached, the $\dot{\alpha}$ reduces to almost zero. The controller holds the angle of attack at 30° for almost 2.5 convective times (from 4 to 6.25 t^*) as annotated in Fig. 11 until the leading-edge vortex breaks down since any further increment in angle of attack would result in a higher C_L error.

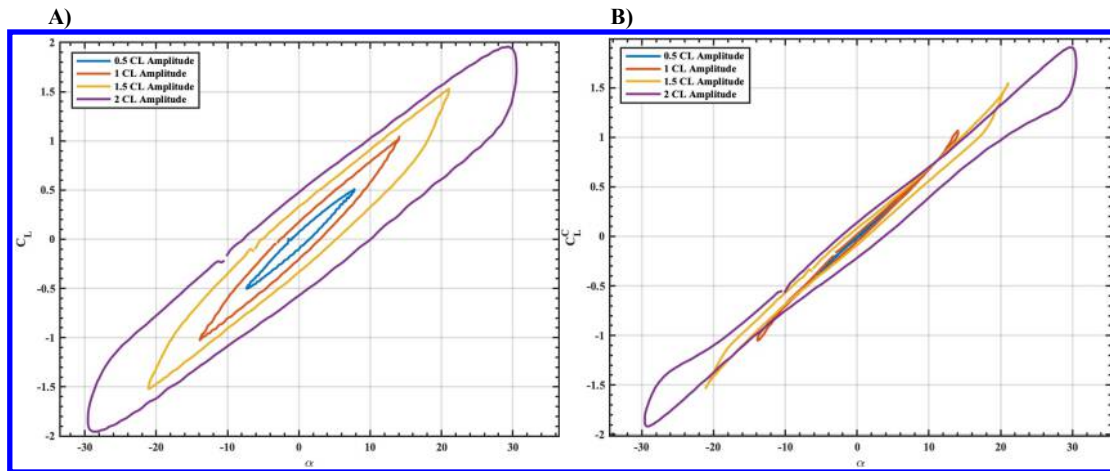


Fig. 10 Closed Loop C_L vs Angle of Attack for $k = 0.2$ (A) and $C_L - C_{L_{AM}}$ vs Angle of Attack (B)

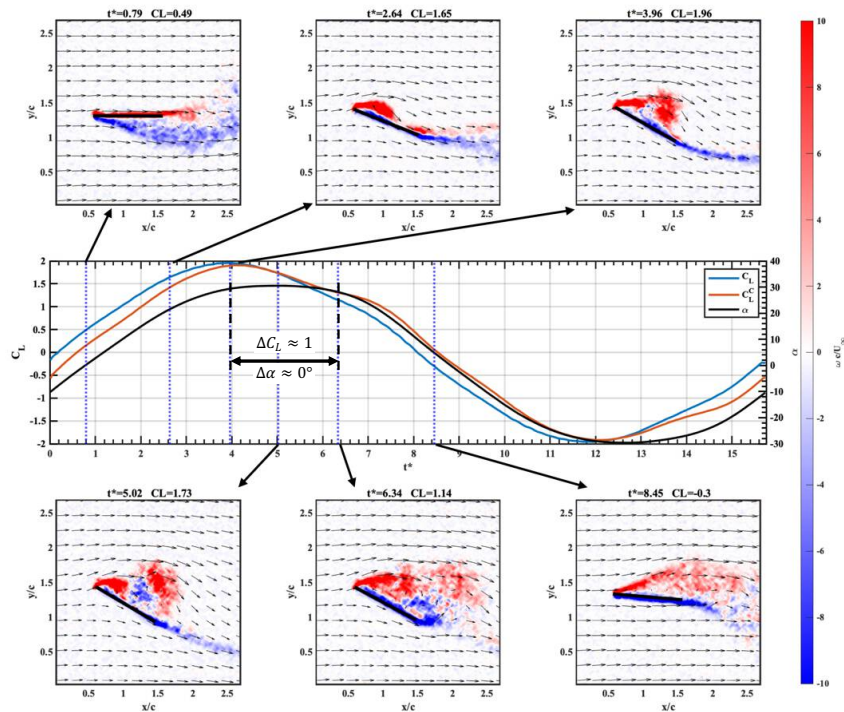


Fig. 11 Flat Plate Closed Loop Profile ($k = 0.2$ $C_{L_{AMP}} = 2$) of C_L , α , and Select PIV Frames

Once the LEV breaks down as shown in Fig. 11 at $6.34t^*$, significant reduction in lift occurs within 1 convective time and the controller starts to respond by initiating the pitch down motion. Similar behavior was observed in the $k = 0.2$ $1.5 C_L$ case as well. The C_L^C component shows that at this reduced frequency the added mass is beginning to have a larger effect where the decreased magnitude almost aligns in phase with the angle of attack. As shown in Table 1, the C_L amplitude of 2.5 case at $k = 0.2$ show poor tracking with 28% error. It is believed that the inability of the controller to achieve the desired C_L amplitude of 2.5 was due to the early LEV bursting when compared to the 2.0 C_L amplitude case. The flow field of this case is shown in Fig. 12 with the vortex breakdown showing up at $6.34 t^*$. This led to a drop in C_L as the plate continued to positively pitch and never achieve the target C_L . This may give rise to an upper limit for the PID controller C_L amplitude where the formation and release of the leading-edge vortex contributes significantly to the achieved C_L . A controller which takes into account the states of the vortex may achieve much better tracking and higher C_L amplitudes than a PID in these cases or a controller that is commanding $\dot{\alpha}$ or $\ddot{\alpha}$ directly in order to control vortex development. Between the $k = 0.1$ and $k = 0.2$ cases it is clear that a similar C_L tracking profile was able to be achieved by the controller through sensing the error and performing adjustments in completely different flow regimes with different flow features.

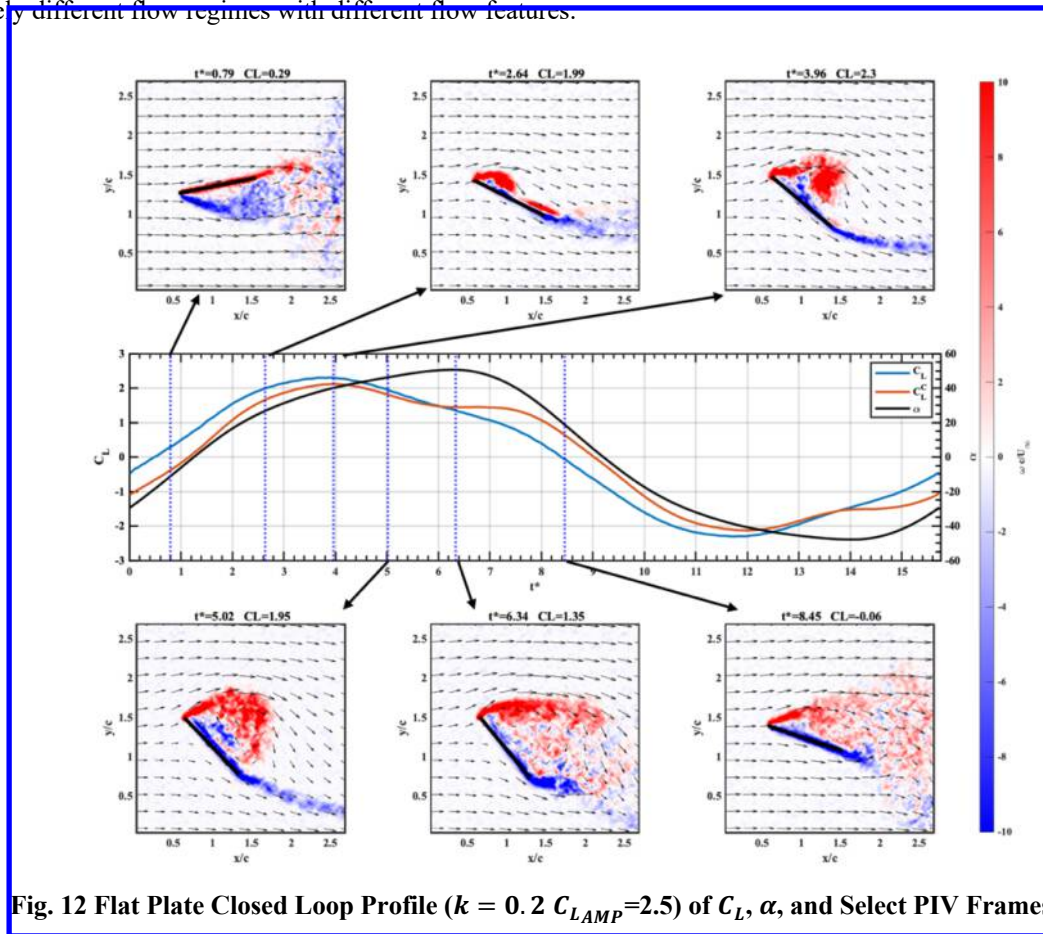


Fig. 12 Flat Plate Closed Loop Profile ($k = 0.2$ $C_{LAMP}=2.5$) of C_L , α , and Select PIV Frames

The results from $k = 0.3$ cases are shown in Fig. 13 for different C_L amplitudes. The shape of the profiles in Fig. 13 (A) are similar to those seen in the $k=0.2$ cases in Fig. 10 (A). However, the transition from pitch up to pitch down and vice-versa doesn't require C_L hold or α hold by the controller. This behavior appears to be the same throughout the various amplitudes. Fig. 13 (B) shows the C_L^C versus α for the $k = 0.3$ cases where a common profile or shape of the lift response appears to be present that scales with increasing C_L amplitude. This is due to the fact that at these higher reduced frequencies, the largest C_L contribution at a given α comes from the added mass component. It should also be noted that the hysteresis at the transition is significantly reduced in Fig. 13 (B) when compared to the lower reduced frequency cases. This is likely due to the formation of a stable LEV that is only shed during the transition as opposed to an unstable LEV that bursts during the pitch up or pitch down maneuver in the case of $0.2 k$.

The lack of a drop off in the lift such as that which occurred in the $k = 0.2$ case is likely due to the stability of the leading-edge vortex as it convects away from the plate. This is apparent in the differences between the behavior of the vortex between Fig. 14 ($k = 0.3$) when compared to Fig. 11 ($k = 0.2$) where over a similar convective time difference (~ 2.5) the vorticity is significantly more contained in the region above the upper surface, indicating a stable LEV in the $k = 0.3$ case than in the $k = 0.2$ case. This smoother movement of the vorticity away from the plate provides a smoother gradient of change in the C_L when compared to the actively broken-down vortex which appears to have a large decremental effect on the C_L . The circulatory C_L component now deviates greatly across the entire time history with added mass clearly having a large effect at this reduced frequency.

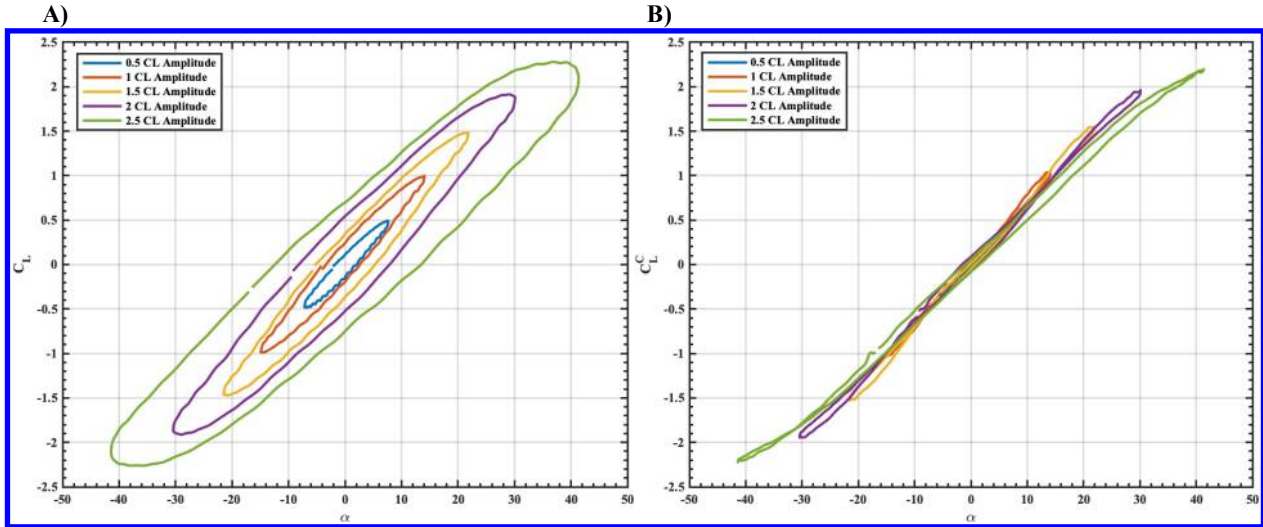


Fig. 13 Closed Loop C_L vs Angle of Attack for $k = 0.3$ (A) and C_L^C vs Angle of Attack (B)

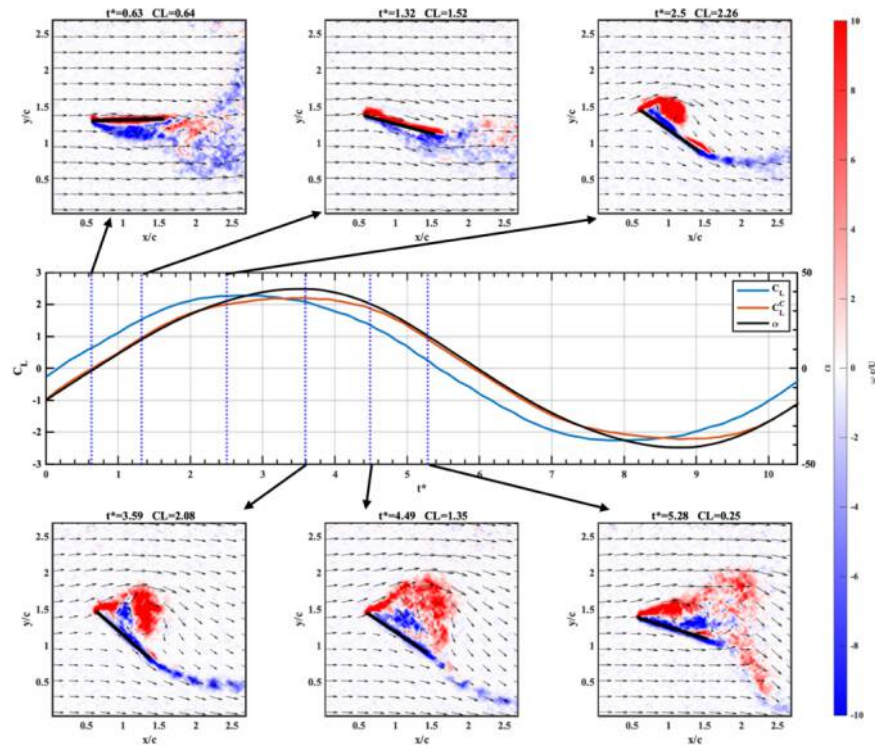


Fig. 14 Flat Plate Closed Loop Profile ($k = 0.3$, $C_{L,AMP} = 2.5$) of C_L , α , and Select PIV Frames

The normalized C_L vs α profiles are shown in Fig. 15 for $k = 0.1$ and $k = 0.2$ as well as Fig. 16 for $k = 0.3$. The normalization was conducted using the maximum C_L and maximum α in each case. The shape that these normalized profiles take for the $k = 0.1$ cases in Fig. 15 (A) changes significantly as the C_L amplitude is increased with the 0.5 C_L case having a slight expansion at low angles of attack and sharp points near the α transition. The 1 and 1.5 C_L cases have larger hysteresis behavior throughout the profile with smoother transitions near the points of α transition due to dynamic stall type behavior.

In the $k=0.2$ cases shown in Fig. 15 (B) the different C_L amplitude cases are much more similar to one another than in the $k=0.1$ cases. The regions at lower angles of attack are more aligned due to the dominant added mass contribution throughout this region when compared to the $k=0.1$ cases. The transition regions show similar behavior except for the C_L 2 amplitude case where a α hold was necessary to maintain the target C_L . This α hold is due to the LEV bursting that caused a sharp change in C_L near the maximum α . The $k = 0.3$ cases shown in Fig. 16 show a significantly aligned set of results with similar shapes throughout the entire profile especially for the C_L amplitudes of 0.5, 1, 1.5, and 2. The C_L amplitude 2.5 case has a slightly larger hysteresis loop throughout but maintains the same overall shape as the others. The shape of the $k = 0.3$ profiles near the maximum α is expectedly rounder than those of the $k = 0.2$ cases due to the stable LEV and shedding that leads to smoother transition. The collapse of the hysteresis loop shows a common trend in aerodynamics around the plate, especially at a higher reduced frequency where stable LEV is formed and shed throughout the motion. However, at lower reduced frequencies, different trends in hysteresis loop show different flow features that the controller had to adjust to maintain C_L tracking. While the tracking error was indeed higher in the higher amplitude cases, the α trajectories that the controller took were different between the C_L amplitude cases at lower reduced frequencies.

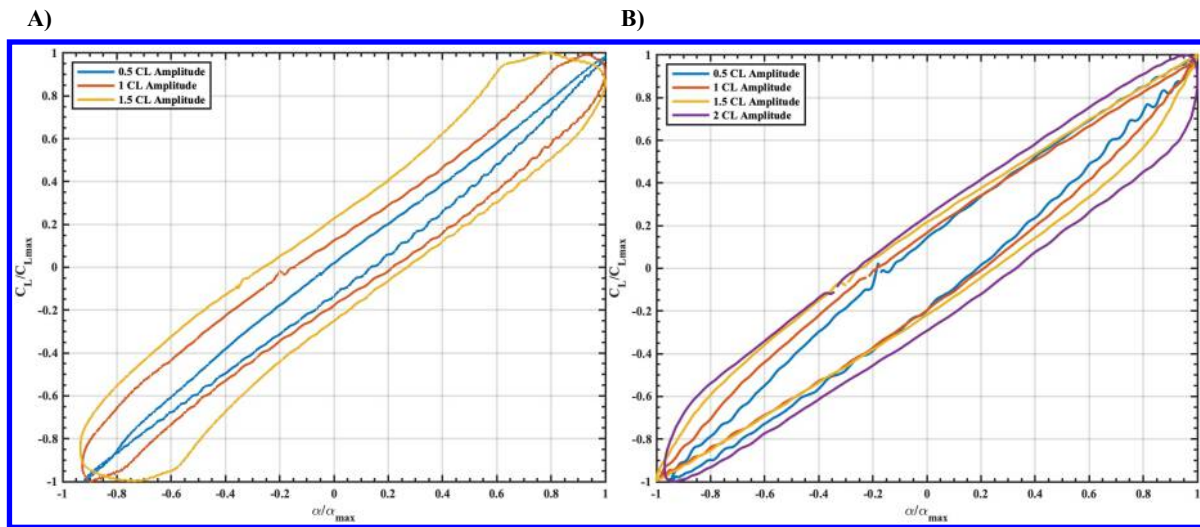


Fig. 15 Normalized C_L vs α for $k = 0.1$ (A) and $k = 0.2$ (B)

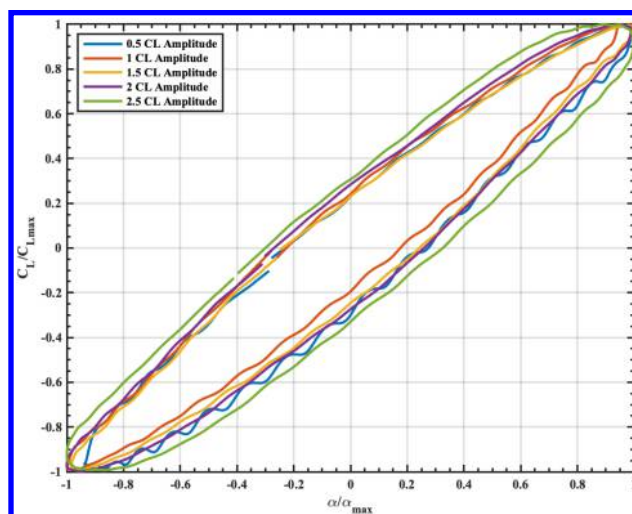


Fig. 16 Normalized C_L vs Normalized α for $k = 0.3$

A similar normalization was done for just the C_L^C component and the flow features around the plate at the maximum C_L^C condition (marked by a black star) is shown in the subplots in Fig. 17 for the $k = 0.1$ case. It is clear that vorticity and velocity vectors seen in the $1 C_L$ and $1.5 C_L$ cases are relatively similar, but a larger region is present in the $1.5 C_L$ case. It is also worth noting that the maximum C_L doesn't always correspond to maximum α . As the C_L amplitude increases, the maximum C_L leads maximum α as shown in Fig. 9. However, for the $k = 0.2$ case shown in Fig. 18, the maximum C_L corresponds to maximum α for almost all C_L amplitude cases that clearly indicates a change in the flow dynamics when compared to $k = 0.1$ case. With the increase in C_L magnitude, a larger leading edge vortex is formed that bursts at different rates, seen as a drastic decrease in C_L for C_L amplitude of 2.0. This is due to the slightly different ways in which the vorticity convects away from the plate as the LEV bursts. In the $k = 0.3$ cases shown in Fig. 19, the circulatory C_L profiles match significantly better due to the sustained presence of the leading-edge vortex (stable LEV) near the plate. The PIV frames again show a common pattern of a building vorticity in the region right above the upper surface with velocity vectors in similar directions

The normalized profiles shown in Fig. 17, Fig. 18, and Fig. 19 cover a breadth of circulatory and added mass contributions at varying levels towards the net C_L which was tracked. These sometimes yielded a similar flow geometry with different scaling but regardless of the uniqueness or similarity in the flow field, the PID controller was capable of tracking a harmonic C_L through regimes of significantly different flow characteristics with highly nonlinear properties and varying circulatory and added mass contributions. The linearization of the system by means of implementing a PID controller has allowed the system to perform adequately and provided valuable insight into the flow field present in these maneuvers.

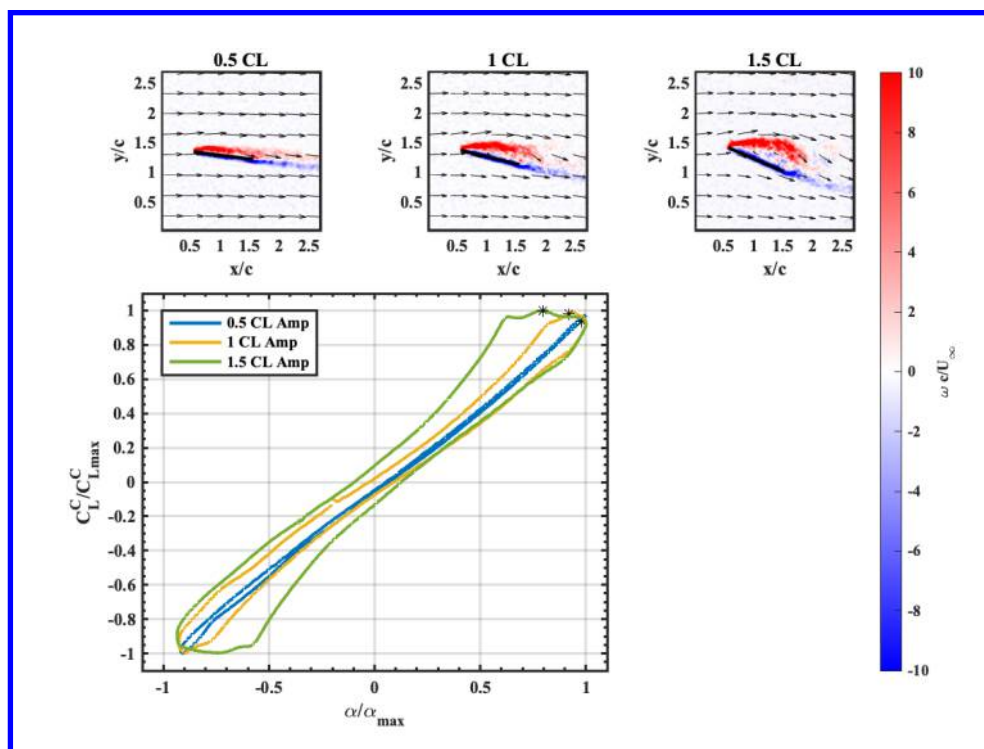


Fig. 17 Normalized Circulatory C_L vs α and Common PIV Frames for $k = 0.1$

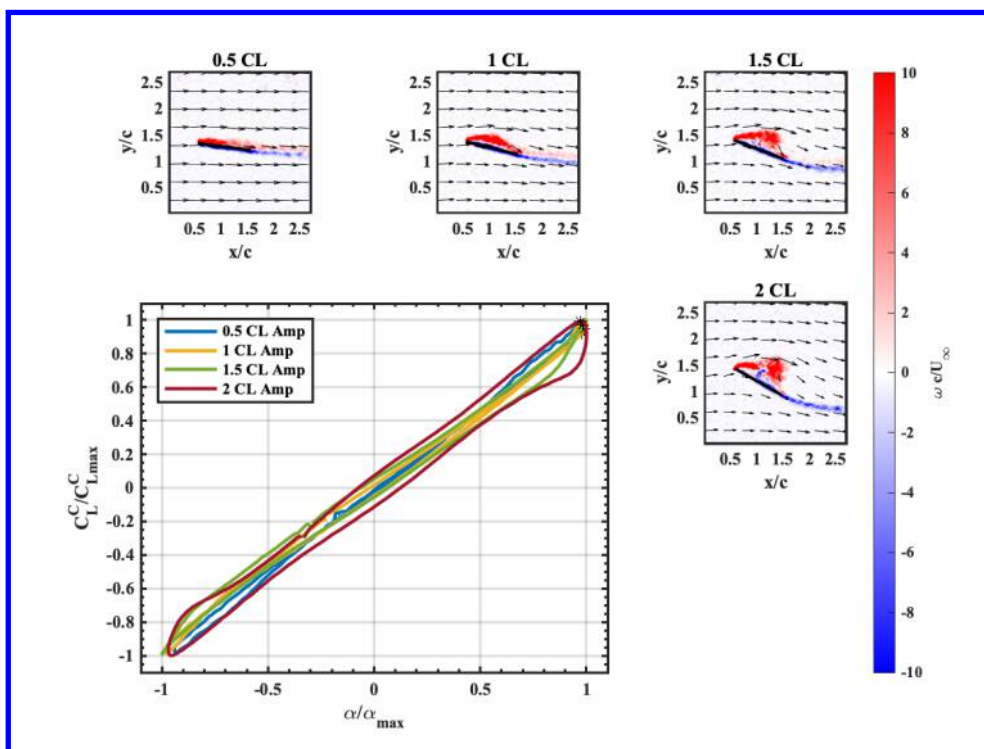


Fig. 18 Normalized Circulatory C_L vs α and PIV Frames for $k = 0.2$

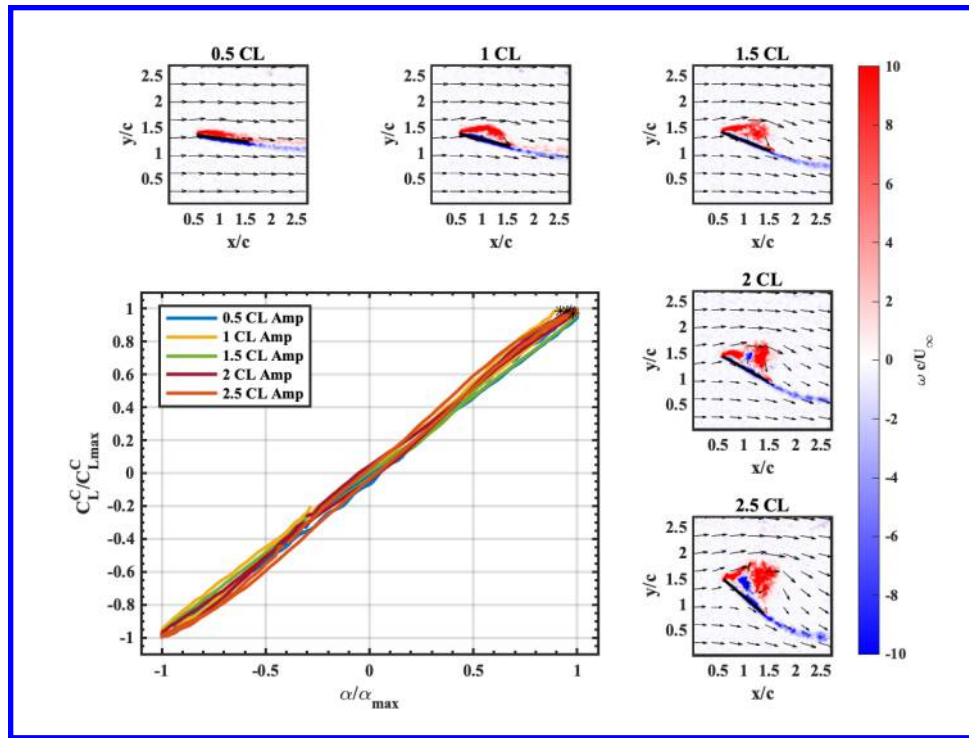


Fig. 19 Normalized Circulatory C_L vs α and PIV Frames for $k = 0.3$

IV. Conclusions

A PID based closed loop controller was used to track a harmonic C_L oscillations across a wide range of reduced frequencies and C_L amplitudes. The tracking error was minimal at lower C_L amplitudes and increased in upwards of 20% at the highest C_L amplitude tested for each reduced frequency case. Active tailoring of the angle of attack by the controller to track the given C_L reference was unique for each case, especially at lower reduced frequencies leading to a non-harmonic change in α . Such non-linearity between C_L and α might be difficult to model in an open-loop setting. The resultant changes in flow field to the controller's response were captured using PIV that showed the presence of different flow features in the three different reduced frequencies tested. The controller was also able to adjust to the varying levels of contribution from added mass and circulatory components towards the net C_L , especially at the lower reduced frequencies tested. The successful tracking of the linear controller in these vastly different unsteady regimes indicates that there are aspects of partial and fully separated flow dynamics that are linear in nature. This work showed that while the flow dynamics were significantly different across the range of $k=0.1$ through $k=0.3$, including the formation of LEV's at the higher frequencies, a PID controller is capable of performing adequately to track a harmonic C_L reference. The implementation of a controller that commands higher order dynamics such as $\dot{\alpha}$ or $\ddot{\alpha}$ might yield a higher bandwidth controller due to its ability to affect the added mass components of lift more directly and tailor the response to LEV bursting more readily.

V. References

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